

数学作业纸

科目_____

班级:_____

姓名:_____

编号:_____

第____页

5.1.1 $V_B > V_C > V_A$ 为共射极。

$\therefore V_C - V_B = -0.2V$ B为发射极, A为集电极, PNP型。

5.2.3. (a) 无级联作用, 无直流偏置。

(b) 有, $V_E > V_B > V_C$ 。

(c) 有, $V_E > V_B > V_C$ 。

(d) 有。

(e) 有。

$$5.2.5 \quad V_{CC} - V_{CEQ} - I_{CQ}(R_C + R_E) = 0.$$

$$R_E = 2.7k\Omega.$$

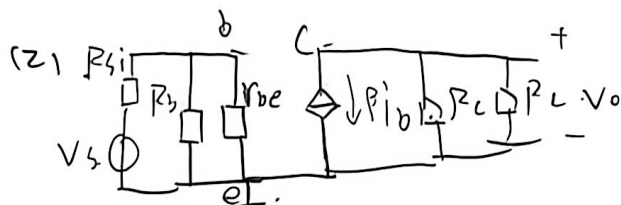
$$I_{BQ} = \frac{I_{CQ}}{100} = 0.01mA.$$

$$I_1 = \frac{V_{CC} - V_{BEQ} - I_{CQ}R_E}{R_{b1}} \quad R_{b1} = \frac{V_{CC} - V_{BEQ} - I_{CQ}R_E}{10 \times I_{BQ}} = 86k\Omega.$$

$$5.2.7. (1) \quad I_{BQ} = \frac{V_{CC} - V_{BEQ} - (-V_{EE})}{R_b} = 40\mu A.$$

$$I_{CQ} = \beta I_{BQ} = 4mA.$$

$$V_{CEQ} = -12V + I_{CQ}R_C = -4V.$$



$$(2) \quad r_{be} = 200\Omega + (1+\beta) \frac{26mV}{I_{CQ}} = 857\Omega.$$

$$A_v = \frac{V_o}{V_i} = -\frac{\beta(R_C \parallel R_L)}{r_{be}} = -155.6.$$

$$R_i = R_b \parallel r_{be} = 0.855k\Omega.$$

$$R_o \approx R_C \parallel R_L = 2k\Omega.$$

数学作业纸

科目_____

班级:_____

姓名:_____

编号:_____

第_____页

(4) 截止失真, 减小 R_b .

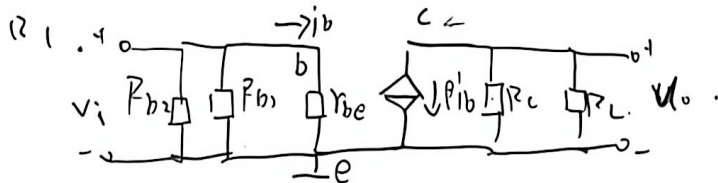
$$5.2.8^{th} V_{BQ} = \frac{R_{b2}}{R_{b1} + R_{b2}} V_{CC} = 4V.$$

$$I_{CQ} = I_{EQ} = \frac{V_{BQ} - V_{BEQ}}{R_e} = 1.65mA.$$

$$I_{BQ} = \frac{I_{CQ}}{\beta} = 28\mu A.$$

$$V_{CEQ} = V_{CC} - I_{CQ}(R_c + R_e) = 7.75V.$$

$$(2) r_{be} = 200 + (1+\beta) \frac{26mV}{I_{EQ}} = 1.16k\Omega.$$



$$A_v = -\frac{\beta(R_c // R_L)}{r_{be}} = -103.$$

$$[4] I_{CQ} = \frac{V_{CC} - V_{CEQ}}{R_c + R_e} = 2.4mA.$$

$$V_{BQ} = V_{BEQ} + V_{CEQ} = I_{CQ}R_e + V_{BEQ} = 5.5V.$$

$$= \frac{R_{b2}}{R_{b1} + R_{b2}} V_{CC}.$$

$$R_{b1} = 38.2k\Omega.$$

$$5.2.11. I_{DQ} = \frac{I_{EQ}}{1+\beta} = 0.01mA.$$

$$I_{CQ} = I_{EQ} = 1mA.$$

$$I_{CQ} = \frac{12 - V_{CEQ}}{R_c} = \frac{V_{CEQ}}{R_L} \Rightarrow V_{CEQ} = 1V.$$

$$I_{R_c} = \frac{12 - V_{CEQ}}{R_c} = 1.1mA \quad I_L = \frac{V_{CEQ}}{R_L} = 0.11mA.$$

$$V_{EQ} = 0 - I_{BQ}R_b - V_{BEQ} = -1.7V.$$

$$V_{CEQ} = 2.7V.$$

数学作业纸

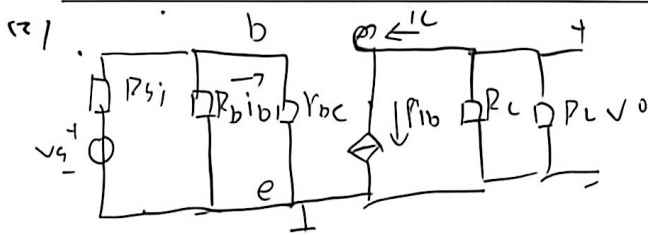
科目_____

班级:_____

姓名:_____

编号:_____

第 页



$$(3) \cdot r_{be} = 200 + (1+\beta) \frac{26}{I_{BQ}} = 2.83 k\Omega$$

$$A_v = \frac{-\beta(R_c \parallel R_L)}{r_{be}} = -176.7$$

$$R_i = R_b \parallel r_{be} = 2.76 k\Omega \quad R_o = R_c = 10 k\Omega$$

5.3.1 (1). b 端输入电阻最大,

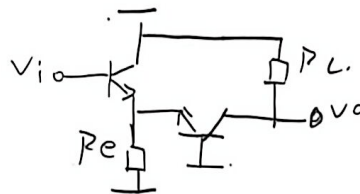
a 端输入电阻最小.

(2) b 端输出电阻最大, c 端输出电阻最小.

(3) c 最小, a 最大.

(4). b 短路.

5.4.2 (1). T_1 为共基, T_2 为共射.



$$(2) \cdot V_{B1Q} = 0 = V_{B2Q}$$

$$V_{E1Q} = V_{E2Q} = 0 - 0.7 = -0.7V$$

$$I_{C1Q} = I_{C2Q} = \frac{I_{RE}}{2} = \frac{-0.7 - (-6)}{2} = 1.33mA$$

$$V_{CE1Q} = V_{CC} - V_{E1Q} = 6.7V$$

$$V_{CE2Q} = V_{CC} - R_C I_{C2Q} - V_{E2Q} = 3.8V$$

$$(1) r_{be1} = r_{be2} = 200 + (1+\beta) \frac{26mV}{I_{C1Q}} = 1.2 k\Omega$$

$$A_{v1} = \frac{(1+\beta)(R_E \parallel \frac{r_{be2}}{1+\beta_2})}{r_{be1} + (1+\beta)(R_E \parallel \frac{r_{be2}}{1+\beta_2})} = 0.15$$

$$A_{v2} = \frac{\beta R_E}{r_{be2}} = 92$$

$$A_v = A_{v1} \cdot A_{v2} = 46$$

$$R_i = r_{be1} + (1+\beta)(R_E \parallel \frac{r_{be2}}{1+\beta_2}) = 2.4 k\Omega$$

$$R_o = R_E = 2.2 k\Omega$$

数学作业纸

科目_____

班级:_____

姓名:_____

编号:_____

第____页

5.4.4 T_1 为基频, T_2 为共集.

$$A_{v1} = -g_m R_{L1} \quad R_{L1} = R_D // [r_{be} + (1+\beta)R_E]$$

$$A_{v2} = \frac{(1+\beta)R_E}{r_{be} + (1+\beta)R_E}$$

$$A_v = A_{v1} \cdot A_{v2} = \frac{-g_m (1+\beta)R_D R_E}{R_D + r_{be} + (1+\beta)R_E}$$

$$R_{I1} = R_g \quad R_{O1} = R_E // \frac{R_D + r_{be}}{1+\beta}$$